# MACHINE-CHECKED RESULTS ON THE MINIMUM NUMBER OF MONOCHROMATIC 3-TERM ARITHMETIC PROGRESSIONS IN TWO-COLOURINGS (ERDŐS PROBLEM 1186, CASE $k = 3$)

CHANDLER SNIVELY

ABSTRACT. Let $\delta_3 = \liminf_{n\to\infty} \min_c V(n,c)/n^2$, where $V(n,c)$ is the number of monochromatic 3-term arithmetic progressions in a 2-colouring $c$ of $\{1,\dots,n\}$ (the case $k = 3$ of Erdős Problem 1186). Parrilo, Robertson and Saracino proved $1675/32768 \le \delta_3 \le 117/2192$ and conjectured that the upper bound is the truth. We prove three statements in Lean 4 with Mathlib, against the definitions of the Formal Conjectures repository, where they are still marked `sorry`. Theorem A is their two bounds. Theorem B is

$$\delta_3 \;\ge\; \frac{14324135}{268435456} = 0.0533615\ldots,$$

within $1.5\cdot10^{-5}$ of $117/2192 = 0.0533759\ldots$, which we did not find in the searches described in Section 9. It combines a sharper cover of the counting region, which recovers the boundary loss of their strip cover (a related cell model appears in Section 7 of the claim discussed below), with a Shor-type semidefinite relaxation with triangle cuts, whose exact integer certificate ($256 \times 256$, 11931 cuts) is checked by the Lean kernel. Theorem C is $\delta_3 = 117/2192$, the conjecture itself: a Lean formalisation, in a discrete form and with five stated departures, of an unrefereed proof claim posted by another author in October 2026 (Section 10). The mathematics of Theorem C is that author's, not ours, and Theorem B is a consequence of it. The mathematics and code behind Theorem B, all the Lean proofs and the text of this paper were produced by AI agents (Claude Opus 5.5 and Claude Sonnet 5.5, Anthropic) in a research workspace directed by the author; Appendix B states their roles.

*Date*: 26 October 2026.

2020 *Mathematics Subject Classification.* 05D10, 11B25, 90C22, 68V15.

*Key words and phrases.* monochromatic arithmetic progressions, Erdős problems, semidefinite programming, triangle inequalities, formal verification, Lean.

**Status of this draft (26 October 2026).** The mathematics of Sections 5–6, all the Lean proofs and the text of this paper were produced by AI agents (Appendix B); the mathematics formalised in Theorem C is another author's [17]. The Lean theorems are checked by the Lean kernel and the certificates also by exact-arithmetic scripts. The prose proofs of Section 5 (the continuum cut-cell lemma and its discrete transfer) are *not* formalised (Lean proves a different, weaker integer argument) and were only checked by symbolic and exhaustive small-case computations and by AI readers; no human mathematician has checked or refereed them. The Lean theorems do not depend on those prose proofs. Reading them as bounds on $\delta_3$ depends on the Lean definitions being the ones of the Formal Conjectures source, which we checked mechanically (Section 7); whether those definitions express the intended constant is the Formal Conjectures authors' claim, not ours. Novelty statements are provisional ("not found in the searches described"). Section 10 reports an unrefereed claim by another author and its Lean formalisation (Theorem C); the argument is not a result of this paper; its author states that they read and checked it line by line, and the claim's report records an outside audit (auditor not named) and a correction of 20 October 2026; we know of no other reading of it apart from that audit and the project's AI reviewers (Section 10), and no human has read the Lean text. The Lean modules were built from source three times: with `lake` in the research repository (for Theorem C, the 131 certificate-side modules after the merge of the branch that contains them, the other 22 as their parts were merged), with plain `lean` from the paper's package (Sections 7.7 and 10.2), and with `lake` from a fresh clone of the release repository, where `lean4checker` also replayed all 342 modules (Section 7.8); every build and every replay succeeded, and an audit of every constant of the 342 modules found only the three standard axioms.

## 1. Introduction

For a 2-colouring $c\colon \{1,\dots,n\} \to \{\pm 1\}$ let $V(n,c)$ be the number of pairs $(a,d)$ with $d \geq 1$ and $a + 2d \leq n$ for which $c(a) = c(a+d) = c(a+2d)$, and let $V_{\min}(n) = \min_c V(n,c)$. Van der Waerden's theorem, with the usual supersaturation argument, gives $V_{\min}(n) \gg n^2$ and a random colouring gives $V_{\min}(n) \leq (\frac{1}{16} + o(1))n^2$. For $k$-term progressions let $\delta_k = \liminf_n V^{(k)}_{\min}(n)/n^2$. Problem 1186 of the Erdős problems database [1], which cites [7, p. 93], asks for reasonable bounds, or even an asymptotic formula, for $\delta_k$; its formalisation in [15] reads this as the order of magnitude of $\delta_k$ as $k \to \infty$ and is open. This paper is only about $k = 3$, the one case where bounds on the constant itself are known. There Parrilo, Robertson and Saracino [11] proved

$$\frac{1675}{32768} = 0.05111\ldots \;\leq\; \delta_3 \;\leq\; \frac{117}{2192} = 0.05337\ldots, \tag{1}$$

the upper bound by an explicit 12-block colouring (found independently by Butler, Costello and Graham [3]; see [4]), and conjectured that it is sharp. According to the database, Graham offered \$100 at a conference in 1999 for the value of $\delta_3$ (the page attributes this report to [11]); no value has been established in the refereed literature (but see Theorem C). Greenwood, Kariv and Williams [8] showed that the 12-block colouring is optimal among anti-symmetric colourings with at most 12 blocks.

This paper has three Lean theorems: Theorem A formalises the known bounds of [11], Theorem B was not found in the searches of Section 9, and Theorem C formalises another author's claim; it also reports some evidence.

**Theorem A** (Lean). *The bounds* (1) *hold:* $1675/32768 \leq \delta_3 \leq 117/2192$, as the statement `erdos_1186.variants.prs_bounds` of Formal Conjectures [15], with the definitions $V$, $V_{\min}$ and $\delta$ copied verbatim from that repository. (That variant is `sorry` there at commit c83a94d,

and still on the main branch on 26 October 2026, as is the main statement `erdos_1186` about $\delta_k$ for all $k$, which we do not touch. We found no earlier formal proof; see Section 9.)

**Theorem B** (Lean). $\delta_3 \geq \dfrac{14324135}{268435456} = 0.05336156\ldots$. Also proved: $\delta_3 \geq 1788837/33554432 = 0.0533114\ldots$ and $\delta_3 \geq 1782409/33554432 = 0.0531199\ldots$ (smaller certificates).

**Theorem C** (Lean; the mathematics is another author's). $\delta_3 = 117/2192$, as the statement `erdos_1186.variants.prs_conjecture` of Formal Conjectures [15] (`research open` and `sorry` there at commit c83a94d), with no hypothesis. It is a Lean formalisation, by AI agents of the research project, of the unrefereed proof claim of Carlos Toledo [17] (posted on the problem's proof-claims thread; produced, by the claimant's own statement, with substantial AI assistance from Claude, and read and checked line by line by the claimant) described in Section 10, in a discrete form with the departures listed in Section 10.1; it implies Theorems A and B. Its Lean proof reuses parts of the development of Theorems A and B (the definitions, the exact identity of Lemma 2.2, the cover inequality of Theorem 5.7 and the upper bound of Theorem A) but none of their certificate tables.

Theorem B gives the lower bound 0.05336 where [11] has 0.05111, so that, if we accept the Lean statement as the problem, $\delta_3 \in [0.0533615, 0.0533760]$ (rounded outwards), an interval of relative width $2.7 \cdot 10^{-4}$; Theorem C closes this interval, by another route. The proof of Theorem B, which is the mathematics of this paper, rests on two ideas. The first (Section 5) is a cover of the counting region by cells that treats the $2L$ cells cut by the boundary with a sharp cut-cell lemma, instead of charging them at worst case as the strip cover of [11] does; it recovers the term $1/(8L)$ that the strip cover loses (a related cell model, which charges cut cells by their areas, appears in Section 7 of the claim; see "Credit and origin" below). The second (Section 6) is to bound the resulting quadratic form on the cube $[-1, 1]^L$ not by a diagonal semidefinite certificate (as [11] do), which for this cover stalls near 0.0522, but by a Shor relaxation with triangle inequalities, whose dual certificate is an integer matrix of size $L \times L$ with thousands of cuts. The numerical checks that the Lean kernel performs are finite integer arithmetic; the reductions around them are ordinary Lean proofs.

**What is and is not ours.** We did not discover the mathematics of $\delta_3 = 117/2192$: the argument is an unrefereed claim by another author, Carlos Toledo [17], and Theorem C says only that it has been formalised in Lean. The evidence of Section 8 (exact certificates (three of them also checked in Lean) leave a gap between $0.94/L^2$ and $1.06/L^2$ to $117/2192$ for $L = 64, 128, 192, 256$) is consistent with the equality but, taken alone, proves nothing about $L \to \infty$. Three AI reviewers in the same research project examined the claim before it was formalised: one reached the verdict "undetermined end to end, with no gap found"; a second rebuilt the claim's five finite problems from its text, in its own reading of the definitions, and certified them with certificates of its own (the claimant's certificates were not run), and a third re-verified those certificates. None of them proved the lemmas of the argument or its counting reduction. Section 10 describes the claim and what has been checked about its Lean formalisation, including what was not checked. Since Theorem B is a consequence of Theorem C, the value of Theorem B is that its proof is a different route, with different mathematics, that does not use the claim's argument.

**Credit and origin.** The lower-bound method of Section 3 and the colouring of Section 4 are those of [11]; Theorem A is a formalisation of known mathematics. The cover of Section 5 with its cut-cell lemma, and the certificates of Sections 6–8, are, in the sense of Section 9, "not found in the searches described"; they were produced by AI agents. Two of the ideas are *not* new to

this paper. A Shor relaxation with triangle inequalities is a standard device (Section 6). And Section 7 of the claim's paper (Section 10) already uses a cell model that charges each cut pair its inside and outside areas instead of the worst case, and bounds it by such a relaxation, with a bound of 0.0532492 at $L = 256$ that the claim describes as certified and that, by the claim's own statement, its second verifier did not re-decide (not reproduced here; Table 5). The claim does not compare that model with the strip cover of [11]; that comparison is ours. The cover of Section 5 was developed on 26 October 2026, after the project had read the claim's report page: the task that asked for it proposed treating the boundary cells exactly and suggested that the claim's cell model might work that way, so the cover was not developed independently of the claim. What we believe is specific to the present cover is the cut-cell estimate of Lemma 5.2 (weights $\frac{1}{8}$ and $\frac{3}{8}$), the retained parity term, the resulting higher minima (floating-point local-search minima at $L = 64, 128, 256$: 0.053031, 0.053267, 0.053347 for the claim's model, as published in the claim and reproduced by a reviewer, against 0.053120, 0.053312, 0.053362 for ours; the comparison is between two float computations and assumes that the two $L$'s are comparable), the exact certificates and the Lean proofs. The mathematics of the claim in Section 10, and therefore of Theorem C, is its author's; the Lean formalisation of it was written by AI agents.

**Plan.** Section 2 fixes the definitions and the exact $\pm 1$ identity. Section 3 proves the lower half of Theorem A with the strip cover; Section 4 the upper half. Sections 5 and 6 prove Theorem B. Section 7 describes the formalisation and what it does and does not cover. Section 8 reports the evidence beyond Lean; Section 9 the prior figures; Section 10 the claim and Theorem C; Section 11 limitations. Appendix A is the reproducibility guide and Appendix B the AI-use statement. Letters are reused between sections ($m$, $K$, $A$, $B$, $\sigma$, $N$ and others); each is defined where it is used.

## 2. Definitions and the exact identity

2.1. **Definitions.** Following [15]: for $k \geq 2$, $n \geq 1$ and $c\colon \mathbb{N} \to \{\text{true}, \text{false}\}$, $V_k(n, c)$ is the cardinality of $\{(a, d) \in [1, n]^2 : \ a + (k-1)d \leq n, \ c(a + id) = c(a) \text{ for all } i < k\}$; $V_{\min}^{(k)}(n) = \min_c V_k(n, c)$ and $\delta_k = \liminf_{n\to\infty} V_{\min}^{(k)}(n)/n^2$. (Only $c|_{[1,n]}$ matters.) We write $V = V_3$, $V_{\min} = V_{\min}^{(3)}$, $\delta = \delta_3$ and use the sign $s_x = \pm 1$ of $c(x)$. The definition agrees with OEIS A121385 [2] for all $n \leq 26$ (exhaustive search, `computation/small-cases/oeis_small.py`).

**Lemma 2.1** (monotonicity)**.** *$V_{\min}(n) \leq V_{\min}(n+1)$. Hence if $L \geq 1$ and $V_{\min}(Lm) \geq \beta L^2 m^2 - aLm - b$ for all $m \geq 1$, then $\delta \geq \beta$.*

*Proof.* Restricting a colouring of $[1, n+1]$ to $[1, n]$ removes progressions only. For the second claim put $n' = L\lfloor n/L \rfloor \geq n - L$; then $V_{\min}(n)/n^2 \geq V_{\min}(n')/n^2 \geq (\beta n'^2 - an' - b)/n^2 \to \beta$. □

2.2. **The $\pm 1$ identity.** Index a progression by its middle term $y$ and difference $d \leq D_y := \min(y - 1, n - y)$. Then there are $N_{AP} = \sum_y D_y = \lfloor (n-1)^2/4 \rfloor$ progressions, and for three signs $4[\text{all equal}] = 1 + s_1 s_2 + s_2 s_3 + s_1 s_3$. Summing over all progressions gives the following.

**Lemma 2.2.** *Let $W = \sum_{(a,b)\in R_n} s_a s_b$ with $R_n = \{(a, b) \in [1, n]^2 : \ 1 \leq 2b - a \leq n\}$, and let $E, O$ be the sums of $s_x$ over even and odd $x \leq n$. Then*

$$4V(n, c) = N_{AP} + (W - n) + \tfrac{1}{2}(E^2 + O^2 - n) \ \geq \ N_{AP} + W + \tfrac{1}{4}(E + O)^2 - \tfrac{3}{2}n,$$

*and consequently $32\,V(n, c) \geq 2n^2 - 16n + 8W + 2(\sum_x s_x)^2$.*

*Proof.* For fixed $y$, the $a$ with $(a, y) \in R_n$ are exactly $a \in [y - D_y, y + D_y]$, so $\sum_y s_y \sum_{1\le|e|\le D_y} s_{y+e} = W - n$ (the diagonal $a = b$ contributes $n$). The terms $s_{y-d}s_{y+d}$ run once over all pairs $x < z$ of equal parity, whose sum is $\frac{1}{2}(E^2 + O^2 - n)$. Finally $E^2 + O^2 \ge \frac{1}{2}(E + O)^2$ and $N_{AP} \ge (n^2 - 2n)/4$. □

Lemma 2.2 is the $\pm1$ form of the reduction of [11, Sec. 2, Lemma 1], which counts the bichromatic pairs $(a, b)$ with $2b - a \in [1, n]$, i.e. over the same region $R_n$; we state it as an exact identity and keep the parity term. The problem becomes a lower bound for $W$, a quadratic form in the $\pm1$ vector $s$ over the *parallelogram* $R_n$.

## 3. The strip cover of Parrilo, Robertson and Saracino

Fix an even $L = 2h$ and $n = Lm$. Let $I_i = ((i - 1)m, im]$ and $X_i = \sum_{x\in I_i} s_x \in [-m, m]$ $(1 \le i \le L)$. Put $B_{ij} = 1$ if $j + 1 \le 2i \le j + L$ and 0 otherwise, and $A = B + B^\top$ (so $A_{ii} = 2$).

**Lemma 3.1** (strip cover)**.** $W \ \ge \ \frac{1}{2}X^\top AX - \frac{1}{2}Lm^2$.

*Proof.* For $y = (i - 1)m + \beta$ $(1 \le \beta \le m)$ the window $\{a : (a, y) \in R_n\}$ is the interval $[\max(1, 2y - n), \min(2y - 1, n)]$. The union of blocks $U_i = \bigcup_{B_{ij}=1} I_j$ is the same interval moved by $2\beta - 1 - m$ at its one non-clipped end (the other end is 1 or $n$ in both). So $\left|\sum_{a\in\text{window}} s_a - \sum_{a\in U_i} s_a\right| \le |2\beta - 1 - m|$, and therefore $s_y \sum_{a\in\text{window}} s_a \ge s_y \sum_{a\in U_i} s_a - |2\beta - 1 - m|$ whatever the sign $s_y$. Summing over $y$ in block $i$ gives $\ge X_i \sum_j B_{ij}X_j - \sum_\beta |2\beta - 1 - m|$, and $\sum_{\beta=1}^m |2\beta - 1 - m| \le m^2/2$. Sum over $i$: $\sum_i X_i \sum_j B_{ij}X_j = X^\top BX = \frac{1}{2}X^\top AX$. □

A *diagonal certificate* for $L$ is a vector $d \in \mathbb{Q}^L_{\ge0}$ with $A + \operatorname{diag}(d)$ positive semidefinite, i.e. $X^\top AX \ge -\sum d_i X_i^2$ for all real $X$.

**Proposition 3.2** (PRS08)**.** *If $d$ is a diagonal certificate for the even $L$, then for $n = Lm$ and every colouring,*

$$V(n, c) \ \ge \ \Big(\frac{1}{16} - \frac{1}{8L} - \frac{\sum_i d_i}{8L^2}\Big)n^2 - \frac{n}{2}, \qquad \textit{hence} \qquad \delta_3 \ge \frac{1}{16} - \frac{1}{8L} - \frac{\sum_i d_i}{8L^2}.$$

*Proof.* By Lemma 2.2 (dropping $(\sum s)^2 \ge 0$) and Lemma 3.1, $32V \ge 2n^2 - 16n + 4X^\top AX - 4Lm^2 \ge 2n^2 - 16n - 4m^2 \sum d_i - 4Lm^2$, using $X_i^2 \le m^2$. Divide by 32 and put $m = n/L$; then apply Lemma 2.1. □

**Theorem 3.3** (Theorem A, lower half)**.** $\delta_3 \ge 1675/32768$. *More precisely, $L = 128$ with a certificate of $\sum d_i = 697849/512$ gives $3430919/67108864 = 0.0511246\ldots$, and $L = 256$ with $\sum d_i = 1394087/256$ gives $6928985/134217728 = 0.0516249\ldots$*

The certificates are exact rational vectors for which $A + \operatorname{diag} d$ is shown positive semidefinite by the Lean kernel (Section 7; the Python checks show positive definiteness by exact elimination). The published certificate of [11] has $\sum d_i = 1364$ for $L = 128$ and gives exactly $1675/32768$; ours is a different vector with a slightly smaller sum, so that $0.0511246 > 0.0511169$. The $L = 256$ value is PRS's method at a larger size; we did not find it in print, it is below the numerical 0.052341 of [13], and we do not claim it as a record. The same method at $L = 512$ gives $55698149/1073741824 = 0.0518729\ldots$ (exact, outside Lean).

## 4. The upper bound by the 12-block colouring

Let $c_1$ colour $[1,548]$ in 12 consecutive blocks of sizes $28,6,28,37,59,116,116,59,37,28,6,28$, alternately true and false, and let $c_K(x) = c_1(\lceil x/K \rceil)$ colour $[1,548K]$. Write $a = KA + r$, $d = KD + s$ with $A, D \geq 0$ and $1 \leq r, s \leq K$. The three terms $a, a+d, a+2d$ lie in the coarse cells $A+1$, $A+D+1+u$, $A+2D+1+v$, where the carries $u = [r+s > K] \in \{0,1\}$ and $v = \lceil (r+2s)/K \rceil - 1 \in \{0,1,2\}$ depend on $(r,s)$ only. So $(a,d)$ is a monochromatic progression of $[1,548K]$ iff the coarse triple lies in $[1,548]$ and is monochromatic for $c_1$. Let $M_{uv}$ be the number of pairs $(A,D)$ with $0 \leq A, D < 548$ for which this holds. A finite computation (done by the Lean kernel, in chunks) gives

$$M_{00} = 16304, \qquad M_{01} = M_{11} = 16028, \qquad M_{12} = 15756.$$

Put $R_{00} = \{(r,s) : r+2s \leq K\}$ and $R_{12} = \{(r,s) : r+2s > 2K\}$ in $[1,K]^2$. Then $(u,v) = (0,0)$ on $R_{00}$, $(1,2)$ on $R_{12}$, and $(u,v) \in \{(0,1),(1,1)\}$ elsewhere. Hence

$$V(548K, c_K) \;\leq\; 16028K^2 + 276|R_{00}| - 272|R_{12}| \;\leq\; 16028K^2 + 4|R_{00}| \;\leq\; 16029K^2,$$

using $|R_{12}| \geq |R_{00}|$ (a short counting argument) and $4|R_{00}| \leq K^2$. Since $16029/548^2 = 117/2192$ and the lim inf is at most the limit along the subsequence $n = 548K$, we get $\delta_3 \leq 117/2192$. (The exact count is $16029K^2 - 274K + (K \bmod 2)$, which we checked by computer only for $K \leq 10$; only the inequality is needed and it is what Lean proves.) This is the colouring of [11].

**Theorem 4.1** (Theorem A)**.** $1675/32768 \leq \delta_3 \leq 117/2192$.

Proof: Theorem 3.3 (with $L = 128$) and the display above.

## 5. A sharper cover of the counting region

Fix even $L \geq 4$ and even $w \geq 2$, $n = Lw$, blocks $I_p = ((p-1)w, pw]$, block sums $X_p = \sum_{x \in I_p} s_x$ and means $x_p = X_p/w \in [-1,1]$. In Lemma 3.1 the window of each $y$ was replaced by a union of blocks, losing the boundary triangles. Here the cells $I_p \times I_q$ ($a \in I_p$, $b \in I_q$) of the region $R_n$ are examined one by one.

**Lemma 5.1** (geometry of the cells)**.** *Put $m = 2q - p - 1$. The cell $I_p \times I_q$ lies entirely inside $R_n$ if $1 \leq m \leq L-2$, and is disjoint from it if $m \leq -2$ or $m \geq L+1$. For the $2L$ remaining cells, write $a = (p-1)w + i$ and $b = (q-1)w + j$ with $1 \leq i, j \leq w$: for $m = 0$ the intersection is $\{i \leq 2j-1\}$ (of size $3w^2/4$), for $m = -1$ it is $\{2j \geq i + w + 1\}$ ($w^2/4$), for $m = L-1$ it is $\{i \geq 2j - w\}$ ($3w^2/4$), for $m = L$ it is $\{i \geq 2j\}$ ($w^2/4$).*

*Proof.* $2b - a = mw + (2j - i)$ with $2 - w \leq 2j - i \leq 2w - 1$. The condition $2b - a \geq 1$ fails everywhere iff $mw + 2w - 1 < 1$, i.e. $m \leq -2$; it holds everywhere iff $mw + 2 - w \geq 1$, i.e. $m \geq 1$; for $m = 0$ it reads $2j - i \geq 1$ and for $m = -1$ it reads $2j - i \geq w + 1$. The condition $2b - a \leq n$ is the mirror statement with $m$ replaced by $m - L$. The two families of boundary cells are disjoint. The sizes are elementary sums ($w$ even). □

Let $K_{pq} \in \{1, \frac{3}{4}, \frac{1}{4}, 0\}$ be the fraction of the cell inside $R_n$; the Lean file uses $K4 = 4K \in \{4,3,1,0\}$. Each block is the $a$-block of exactly two cut cells and the $b$-block of exactly two (as an $a$-block, the parity of $p$ decides which two: $m \in \{0, L\}$ for odd $p$, $m \in \{-1, L-1\}$ for even $p$; as a $b$-block, $q \leq L/2$ gives $m \in \{0, -1\}$ and $q > L/2$ gives $m \in \{L-1, L\}$).

5.1. **The cut-cell lemma.** For each cut cell we need a lower bound for $\sum_{(i,j)\in U} s_i t_j$ ($s, t$ the sign vectors of the two blocks, $U$ the part inside $R_n$) in terms of the block means $x, y$ only. This is where the cover differs from the strip cover.

**Lemma 5.2** (cut-cell lemma, continuum form)**.** *Let $\varphi, \psi\colon [0,1] \to [-1,1]$ be measurable, $x = \int \varphi$, $y = \int \psi$, $\Psi(t) = \int_0^t \psi$ and*

$$E(\varphi,\psi) = \int_0^1 \varphi(u)\big[\Psi(u/2) - \tfrac{y}{4}\big]\,du, \qquad \Gamma(x,y) = \frac{1-x^2}{8} + \frac{3(1-y^2)}{8}.$$

*Then $E(\varphi,\psi) \le \Gamma(x,y)$.*

*Proof.* $E(-\varphi,-\psi) = E(\varphi,\psi)$ and $\Gamma$ is even in both arguments, so assume $y \ge 0$. For every constant $c$, since $|\varphi| \le 1$ and $\int \varphi = x$,

$$E = \int \varphi\,(\Psi(u/2) - c)\,du + (c - \tfrac{y}{4})x \;\le\; (c - \tfrac{y}{4})x + 2\int_0^{1/2} |\Psi(t) - c|\,dt.$$

Pointwise $\Psi_{lo}(t) := \max(-t, t+y-1) \le \Psi(t) \le t$ on $[0,\frac12]$ (use $|\psi| \le 1$ on $[0,t]$ and on $[t,1]$), hence $|\Psi(t) - c| \le g(t) := \max(t - c,\, c - \Psi_{lo}(t))$.

*Case 1: $x \ge 2y - 1$.* Take $c = 0$. Then $-\Psi_{lo}(t) = \min(t, 1-y-t) \le t$, so $g(t) = t$ and $E \le \frac14 - \frac{xy}{4}$. Moreover $8\big(\Gamma - \frac{1-xy}{4}\big) = 2 - (x-y)^2 - 2y^2$, and on the triangle $\{-1 \le x \le 1,\ 0 \le y \le \frac{1+x}{2}\}$ the convex function $(x-y)^2 + 2y^2$ is at most its largest vertex value, which is 2 (the vertices $(-1,0), (1,0), (1,1)$ give $1, 1, 2$). So $E \le \Gamma$.

*Case 2: $x < 2y - 1$* (so $y > \frac{1+x}{2}$). Take $c = \frac{y}{2} - \frac{1+x}{4} \in (0, \frac{y}{2}]$, $\tau = \frac{1-y}{2}$ and $t^* = \frac{1-x}{4} \in (\tau, \frac12]$. One checks that $g = c + t$ on $[0,\tau]$, $g = c - t - y + 1$ on $[\tau, t^*]$ and $g = t - c$ on $[t^*, \frac12]$, and integrating,

$$(c - \tfrac{y}{4})x + 2\int_0^{1/2} g = H(x,y),$$
$$H(x,y) := \tfrac18\big(1 + 4y - 4y^2 - 2x + 2xy - x^2\big),$$
$$\Gamma(x,y) - H(x,y) = \tfrac18(1-y)(2x - y + 3) \;\ge\; 0.$$

□

The symbolic identities in the proof are re-verified by `computation/small-cases/lemma2_symbolic.py` (sympy, with a rational-grid check of the piecewise form of $g$). The weights $(\frac18, \frac38)$ cannot be exchanged: at $y = \pm 1$ one has $E_{\max} = (1-x^2)/8 = \Gamma(x, \pm 1)$ exactly, while for $x = -1$ and $y \to 1$ the maximum of $E$ is $\Gamma - (1-y)^2/8$, so the second weight $\frac38$ is also needed; with the weights swapped the inequality fails (at $(x,y) = (-1, 0.9)$ the maximum of $E$ is 0.07, the swapped bound is 0.02375).

**Corollary 5.3** (cut cells)**.** *With $A = \{u \le 2v\}$ and $B = \{2v \ge u+1\}$ in $[0,1]^2$: $\iint_A \varphi\psi - \frac34 xy \ge -\Gamma(x,y)$ and $\iint_B \varphi\psi - \frac14 xy \ge -\Gamma(x,y)$; the cut cells at the other end of $R_n$ are the images of these under $(u,v) \mapsto (1-u, 1-v)$.*

(Fubini gives $\iint_A \varphi\psi = xy - \int \varphi(u)\Psi(u/2)\,du = \frac34 xy - E(\varphi,\psi)$. For $B$, the substitution $(u,v) \mapsto (1-u, 1-v)$ turns it into $\{v \le u/2\}$, so $\iint_B \varphi\psi = \frac14 xy + E(\tilde\varphi, \tilde\psi)$ with $\tilde\varphi(u) = \varphi(1-u)$, and $E(\tilde\varphi,\tilde\psi) = -E(-\tilde\varphi, \tilde\psi) \ge -\Gamma$.)

**Lemma 5.4** (discrete cut-cell inequality)**.** *Let $w \ge 1$ and let $s, t \in [-1,1]^w$ be real vectors, $X = \sum_i s_i$, $Y = \sum_j t_j$, $T(k) = \sum_{j \le k} t_j$ and $D = \sum_{i=1}^w s_i T(\lfloor i/2 \rfloor)$. Then*

$$8D - 2XY \;\le\; (w^2 - X^2) + 3(w^2 - Y^2) + 4w.$$

*Proof.* Let $\varphi,\psi\colon [0,1]\to[-1,1]$ be the step functions equal to $s_i$ on $(\frac{i-1}{w},\frac{i}{w}]$ and to $t_j$ on $(\frac{j-1}{w},\frac{j}{w}]$. Then $x=\int\varphi = X/w$, $y=\int\psi = Y/w$ and $\Psi(k/w)=T(k)/w$ for integers $k$. For $u\in(\frac{i-1}{w},\frac{i}{w}]$ the point $u/2$ is within $\frac{1}{2w}$ of $\lfloor i/2\rfloor/w$ (for $i=2k$ it lies in $(\frac{2k-1}{2w},\frac{k}{w}]$, for $i=2k+1$ in $(\frac{k}{w},\frac{k+1/2}{w}]$), and $|\psi|\le 1$, so $|\Psi(u/2)-T(\lfloor i/2\rfloor)/w|\le\frac{1}{2w}$. Hence

$$E(\varphi,\psi)=\sum_i s_i\int_{(i-1)/w}^{i/w}\Psi(u/2)\,du-\tfrac{xy}{4}\ \ge\ \frac{D}{w^2}-w\cdot\frac{1}{w}\cdot\frac{1}{2w}-\frac{XY}{4w^2}.$$

By Lemma 5.2, $E\le\Gamma(x,y)$; multiply by $8w^2$ and use $8w^2\Gamma(X/w,Y/w)=(w^2-X^2)+3(w^2-Y^2)$. □

**Corollary 5.5** (cut cells, discrete)**.** *Let $s,t$ be as above, with $w$ even, and write $G=\frac{1}{8}[(w^2-X^2)+3(w^2-Y^2)]=w^2\Gamma(X/w,Y/w)$. By Lemma 5.1 the part of the block pair inside $R_n$ is, for the four kinds of cut cell, one of the following sets, and*

$$m=0:\quad\sum_{i\le 2j-1}s_it_j-\tfrac{3}{4}XY=\tfrac{1}{4}XY-D\ \ge\ -G-\tfrac{w}{2},$$
$$m=-1:\quad\sum_{2j\ge i+w+1}s_it_j-\tfrac{1}{4}XY\ge -G-\tfrac{w}{2},$$

*while the cells $m=L-1$ and $m=L$ are the images of these under $(i,j)\mapsto(w+1-i,w+1-j)$ and satisfy the same bounds.*

*Proof.* For $m=0$ the $j$ with $2j-1\ge i$ are those with $j>\lfloor i/2\rfloor$, so the sum is $\sum_i s_i(Y-T(\lfloor i/2\rfloor))=XY-D$; apply Lemma 5.4. For $m=-1$ the substitution $(i,j)\mapsto(w+1-i,w+1-j)$ turns $2j\ge i+w+1$ into $i'\ge 2j'$, i.e. $j'\le\lfloor i'/2\rfloor$, so the sum is $\tilde D$ for the reflected vectors $\tilde s,\tilde t$ (reflection preserves the sums $X,Y$). Apply Lemma 5.4 to $(-\tilde s,\tilde t)$: this turns $D$ into $-\tilde D$ and $X$ into $-X$, and leaves the right-hand side unchanged. The cells $m=L-1$ and $m=L$ are reflected copies of $m=0$ and $m=-1$ (Lemma 5.1), and the same substitution applies in both coordinates. □

*Remark* 5.6 (what Lean proves). The Lean development does not use integrals. It proves the integer inequality `lemA`, which is Lemma 5.4 for even $w$ with the weaker additive error $4w+16$, for integer-valued $s,t$, by the two cases of the proof of Lemma 5.2 carried out in integer arithmetic, together with the reflections (`DC_lower`, `DC_upper`). (For fixed $w$ the left side of Lemma 5.4 minus its right side is convex in $s$ and in $t$ separately, so its maximum over the cube is attained at vertices; an exhaustive check over $s,t\in\{\pm1\}^w$ for $w=2,4,\dots,10$ (`computation/small-cases/disc_cutcell_check.py`) finds a largest excess of exactly 4 in each case. This shows how weak $4w$ and $4w+16$ are, and is not used in any proof.) Theorem B depends on the Lean statement only; Lemma 5.4 is the human-readable argument for the same estimate.

5.2. **The cover inequality and the cube bound.** Let $K4_{pq}=4K_{pq}$, $C=\frac{1}{8}\big(K4+K4^{\top}+8I+2J\big)$ with $J$ the all-ones matrix, and for $x\in\mathbb{R}^L$ write $\mathrm{cform}(x)=x^{\top}Cx$.

**Theorem 5.7** (cover inequality)**.** *For even $L\ge 4$, even $w\ge 2$, $n=Lw$ and every colouring (indeed every integer-valued $s$ with $|s|\le 1$), with $X_p$ the block sums,*

$$8W\ \ge\ 2\sum_{p,q}K4_{pq}X_pX_q-8\sum_p(w^2-X_p^2)-L(8w+32).$$

*Consequently* $32V(n,c)\ge 2n^2-16n-8Lw^2-L(8w+32)+8w^2\,\mathrm{cform}(X/w)$.

*Proof.* $W$ is the sum over all cells $I_p \times I_q$ of the part of $\sum s_a s_b$ inside $R_n$. By Lemma 5.1 a full cell ($K4_{pq} = 4$) contributes exactly $X_pX_q = \frac{K4_{pq}}{4} X_pX_q$, and an empty cell contributes 0. A cut cell with $K4_{pq} \in \{3, 1\}$ contributes at least $\frac{K4_{pq}}{4} X_pX_q - G_{pq} - \frac{w}{2}$, with $G_{pq} = \frac{1}{8}[(w^2 - X_p^2) + 3(w^2 - X_q^2)]$, by Corollary 5.5 (here $s, t$ are the signs of the blocks $p$ and $q$; the Lean statement has the error $\frac{4w+16}{8}$ in place of $\frac{w}{2}$, which is the weaker constant used in the display). Summing over the $2L$ cut cells, $W \geq \frac{1}{4}\sum_{p,q} K4_{pq}X_pX_q - \sum_{\text{cut}} G_{pq} - 2L \cdot \frac{4w+16}{8}$. Each block is the $a$-block of exactly two cut cells and the $b$-block of exactly two, so in $8\sum_{\text{cut}} G_{pq}$ the block $p$ appears with $2{\cdot}1+2{\cdot}3 = 8$ times $(w^2 - X_p^2)$, which gives the display. For the consequence insert this into Lemma 2.2, and use $\sum_{p,q}(8C)_{pq}X_pX_q = 2\sum K4_{pq}X_pX_q + 8\sum X_p^2 + 2(\sum X_p)^2$, in which $2(\sum X_p)^2 = 2(\sum s)^2$ is exactly the term retained in Lemma 2.2. □

Say that $\sigma \geq 0$ is a *cube bound* for $L$ if $x^\top Cx \geq -\sigma$ for all $x \in [-1, 1]^L$.

**Corollary 5.8** (cube bound $\Rightarrow \delta_3$)**.** *If $\sigma$ is a cube bound for the even $L \geq 4$, then for every even $w \geq 2$ and $n = Lw$, $V(n, c) \geq b(L, \sigma)n^2 - n - L$ for all $c$, and*

$$\delta_3 \;\geq\; b(L, \sigma) := \frac{1}{16} - \frac{L + \sigma}{4L^2}.$$

*Proof.* Put $x = X/w \in [-1, 1]^L$ in Theorem 5.7: $32V \geq 2n^2 - 16n - 8n^2/L - 8n - 32L - 8n^2\sigma/L^2$. Divide by 32, use $24n/32 \leq n$ and Lemma 2.1, applied with $2L$ in place of $L$ and $m = w/2$ (so $n = 2Lm$ ranges over the multiples of $2L$). □

*Remark* 5.9 (why it is sharper). With a diagonal certificate for $C$ ($e \geq 0$, $C + \operatorname{diag} e \succeq 0$, $\sigma = \sum e$) this cover gives the exact values 0.0521658 at $L = 128$ and 0.0521505 at $L = 256$ (rounded down), against 0.0511246 and 0.0516249 for the strip cover at the same $L$. The gain, +0.00104 and +0.00053, is about the size of the strip cover's lost term $1/(8L) = 0.00098$, 0.00049. But the diagonal relaxation is far from the true minimum of the quadratic form on the cube (a floating-point multi-start local search at $L = 256$ finds the value 0.0533617 for this cover), and its value stays near 0.0522 for every size computed. The rest of the improvement comes from a better lower bound for $\min_{x \in [-1,1]^L} x^\top Cx$.

## 6. Triangle cuts, exact certificates and Theorem B

6.1. **A relaxation of the cube minimum.** For $x \in [-1, 1]^L$ the matrix $xx^\top$ is positive semidefinite with diagonal $\leq 1$ (Shor's relaxation [12]). For distinct $i, j, k$ and signs $\epsilon_1, \epsilon_2, \epsilon_3 \in \{\pm 1\}$ with product $+1$ (four patterns) the *triangle inequality*

$$\epsilon_1 x_i x_j + \epsilon_2 x_i x_k + \epsilon_3 x_j x_k \;\geq\; -1 \tag{2}$$

holds on the cube (cf. [10]): the left side is multilinear, so it suffices to check the vertices. Patterns with product $-1$ are invalid. (In Lean this is `cutval_ge`.)

**Lemma 6.1** (weak duality)**.** *Let $d \in \mathbb{R}^L_{\geq 0}$ and $\mu_t \geq 0$ for a finite family $t$ of triples with sign patterns, and let $C_t$ be the symmetric matrix with entries $\epsilon_1/2$ at the positions $(i, j), (j, i)$, $\epsilon_2/2$ at $(i, k), (k, i)$ and $\epsilon_3/2$ at $(j, k), (k, j)$ of the triple $(i, j, k)$ with signs $\epsilon$. If $M = C + \operatorname{diag}(d) - \sum_t \mu_t C_t$ is positive semidefinite, then $x^\top Cx \geq -(\sum_i d_i + \sum_t \mu_t) =: -\sigma$ on $[-1, 1]^L$.*

*Proof.* $x^\top Cx = x^\top Mx - \sum_i d_i x_i^2 + \sum_t \mu_t\, x^\top C_t x \geq 0 - \sum d_i - \sum \mu_t$, since $x^\top C_t x$ is the left side of (2) and $x_i^2 \leq 1$. □

6.2. **Integer certificates.** Fix an integer $Q \geq 1$ (we use $Q = 4096$), integers $dn_i \geq 0$ and $mn_t \geq 0$, and put

$$N := 2Q \cdot (8C) + 16\,\mathrm{diag}(dn) - 8\sum_t mn_t\,\Sigma_t,$$

where $\Sigma_t$ carries the signs of the cut $t$ on its six off-diagonal positions. Then $N = 16Q\,M$ for $d = dn/Q$, $\mu = mn/Q$, so if $N \succeq 0$ then $\sigma = (\sum dn + \sum mn)/Q$ is a cube bound. To prove $N \succeq 0$ in exact arithmetic we use an *approximate Cholesky factorisation with diagonally dominant remainder*: an integer $S > 0$ and an integer matrix $G$ such that $E = SN - GG^{\top}$ is symmetric with $E_{ii} \geq \sum_{j \neq i} |E_{ij}|$ for every $i$. Then $S\,x^{\top}Nx = \|G^{\top}x\|^2 + x^{\top}Ex \geq 0$ for all real $x$. (Outside Lean we also use fraction-free Bareiss elimination, all leading principal minors positive, as a second check.)

6.3. **Theorem B.** The multipliers are found by floating-point optimisation (Burer–Monteiro with an augmented Lagrangian, cutting planes, and a repair step) and then rounded to multiples of $1/Q$; this affects only how good a certificate is, never whether it is valid, because validity is the exact check of a finite integer object.

| $L$ | cuts | $\sigma$ | $b(L,\sigma)$ | decimal |
|---|---|---|---|---|
| 64 | 845 | 183671/2048 | 1782409/33554432 | 0.0531199… |
| 128 | 3586 | 242779/512 | 1788837/33554432 | 0.0533114… |
| 256 | 11931 | 2190937/1024 | 14324135/268435456 | 0.0533615… |

TABLE 1. The certificates of Theorem B, all checked by the Lean kernel. $117/2192 = 0.0533759\ldots$

**Theorem 6.2** (Theorem B)**.** *$\delta_3 \geq 14324135/268435456$, and $\delta_3 \geq 1788837/33554432$, $\delta_3 \geq 1782409/33554432$.*

*Proof.* Corollary 5.8 with the cube bounds of Table 1, which follow from Lemma 6.1 and the integer certificates; positive semidefiniteness of $N$ in each case is established by the kernel (Section 7). □

## 7. The formalisation

7.1. **What is in Lean.** The development is a project in Lean 4.27.0 [6] over Mathlib (v4.27.0) [16], 189 modules and 12.6 MB of source in the import closure of the two top files `P1186LowMain` (Theorem A, and the $L = 256$ strip bound) and `P1186M2Main` (Theorem B). Theorem C (Section 10.1) adds 153 modules (8.8 MB), so that the shipped Lean project has 342 modules and 21.3 MB; the module table of this section is for the 189. Table 2 lists the roles. The generic statements are `lower_of_cert` (Proposition 3.2, for every even $L$ and every diagonal certificate), `lower_of_cubeLB` (Corollary 5.8, for every even $L \geq 4$ and every cube bound), `cubeLB_of_cert` (Lemma 6.1 with the kernel-checkable conditions); the instances are `cert128`, `cert256`, `cubeLB64`, `cubeLB128`, `cubeLB256`, and the top-level statements `erdos_1186.variants.prs_bounds`, `delta3_ge_L256`, `delta3_ge_model2_L64`, `delta3_ge_model2_L128`, `delta3_ge_model2_L256`, `delta3_bounds_model2_L256`.

| Module(s) | Content |
|---|---|
| `P1186` | definitions (verbatim from [15]); the upper bound $\delta_3 \leq 117/2192$; also `probabilistic` and `van_der_waerden`, not used in this paper's claims |
| `P1186Tab` | the four constants $16304, 16028, 16028, 15756$ by `decide +kernel` |
| `P1186Prs`, `P1186Low`, `P1186LowA` | strip cover, Proposition 3.2 (any even $L$, any diagonal certificate) |
| `P1186Cert128*`, `P1186Cert256*` | diagonal certificates $L = 128, 256$ (11 and 59 modules: a top module, a table, and 8 resp. 57 chunk modules; the $L = 128$ family also holds the library used by both) |
| `P1186LowMain` | Theorem A, and $\delta_3 \geq 6928985/134217728$ |
| `P1186M2`, `M2Low`, `M2LowA`, `M2LowB` | cell kernel, discrete cut-cell lemma, cover inequality, Corollary 5.8 |
| `P1186M2CertLib` | Lemma 6.1 and the soundness of the integer certificate (generic, no numbers) |
| `P1186M2Cert64*`, `Cert128*`, `Cert256*` | generated tables and chunked kernel checks: 10, 21 and 76 modules, each family counted with its top module |
| `P1186M2Main` | Theorem B |

TABLE 2. Module families in the closure. The generated modules contain no hand-typed numbers; their generator reads the JSON certificates.

7.2. **Statement fidelity.** The definitions of `monoCount`, `minMonoCount` and `delta` are copied verbatim from the pinned upstream file (Formal Conjectures, commit c83a94d785d0, 2026-10-22), differing only in the namespace (`Erdos1186Day` for `Erdos1186`). The Lean theorem `erdos_1186.variants.prs_bounds` has the identical statement $\frac{1675}{32768} \leq \delta\, 3 \wedge \delta\, 3 \leq \frac{117}{2192}$. This is checked mechanically, after whitespace normalisation, by `verification/fidelity_check.py` (log ends `RESULT: ALL OK`), which also records that at the pin the upstream statements `prs_bounds`, `prs_conjecture`, `probabilistic`, `van_der_waerden` and `erdos_1186` are `sorry`. The same script, given `P1186ExFinal.lean` as a fifth argument, checks that Theorem C has the name, the type and the absence of binders of the upstream statement `erdos_1186.variants.prs_conjecture`, and the namespace of the development (`Erdos1186Day`, as for the definitions) (Section 10.2; the log was regenerated and again ends `RESULT: ALL OK`). The upstream docstring says each progression $(a, d)$, $d \geq 1$, is counted once, which is what the definition does; it matches the first 26 terms of A121385 by exhaustive computation (`computation/small-cases/oeis_small.py`). The statement of Theorem B, $\frac{14324135}{268435456} \leq \delta\, 3$, is a statement about the same `delta`. (Some comments in the shipped Lean files give decimals rounded to the nearest digit, for example 0.0533616; the rational in the statement is what is proved, and the paper's decimals of certified lower bounds are rounded down.)

7.3. **Axioms.** For every theorem cited above, and for the instances `cert128` and `cert256`, `#print axioms` prints exactly `[propext, Classical.choice, Quot.sound]`. The audit file is `earlier-runs/Audit1186.lean` (`earlier-runs/` stands for `verification/earlier-runs/2026-10-26-package/` of the release repository, Appendix A); its output was obtained twice: in the research repository (`lean_axiom_audit.log`,

which does not print `cert128`, `cert256`) and on the shipped tree built from scratch (`lean_axiom_audit_shipped_tree.log`, Section 7.7). The audit of Theorem C and its parts is the separate file `earlier-runs/AuditClaim1186.lean` (Section 10.2). In the release repository, `verification/Headline.lean` prints the statements and axioms of the 36 theorems this paper uses, and `verification/AxiomAudit.lean` the axioms of every constant of the 342 modules (Section 7.8). No `sorry`, `native_decide` or additional axiom is used; a source scan of the package listed every line of the closure containing a forbidden word (log `earlier-runs/closure_scan_shipped_tree.log`), the release repository's `verification/source_scan.py`, which removes comments, docstrings and strings first, finds none in the 342 modules, and the authoritative test is the `#print axioms` output.

7.4. **Kernel checks.** All certificate checks use `decide +kernel`. For the cut certificates the kernel checks: (1) every cut is well formed, $dn \geq 0$, $mn \geq 0$, and $48 \sum mn < 2^{30}$; (2) an accumulation pass that packs the cut part of every row of $N$ into one integer in base $2^{32}$; (3) for each row $a$, that the table row $N_a$ equals $2Q(8C)_a + 16dn_a - 8$(cut part): the cut part is never summed entrywise, because an identity between packed integers together with a bound on each digit forces the identity digit by digit; (4) the dominance condition of the approximate Cholesky factorisation, row by row. The generic soundness theorems (`Cert.link_sound`, `Cert.psd`, `cubeLB_of_cert`) are hand-written and contain no numbers. Technical constraints shaped the layout: one `def` per row (a single list literal for all rows times out), chunks of 400 ($L = 128$) or 800 ($L = 256$) cuts (a `List.foldl` over more than about 1000 items exceeds the kernel recursion depth), and about 90,000 multiply-adds per module because memory is not returned between declarations. The gated build times reported by the formalising agents were 282 s ($L = 64$), 762 s ($L = 128$) and 5426 s ($L = 256$, 10.2 MB of generated source); these runs were under a 9 GB memory limit, and peak memory was not measured. **Negative controls.** According to the formalising lane's report (no log of this is shipped; the script is, and the controls were not re-run, nor in the run of record of Section 7.8), seven corruptions of an $L = 32$ certificate are each rejected by the kernel at the expected check: one entry of a Cholesky row; one entry of $N$; one multiplier $mn$ raised by one with the accumulators and $N$ untouched; an intermediate accumulator; every $dn$ scaled by 0.99 with $N$ untouched; one sign of one cut flipped (product $-1$); and a mutually consistent corruption (one multiplier raised and all accumulators recomputed), which shows that the link between $N$ and the cut list is what binds them (`computation/theorem-b/lean_cert2_controls.py`).

7.5. **What the formalisation does not cover.** (i) The continuum Lemma 5.2 and the transfer to the discrete Lemma 5.4 (the written proof of the discrete estimate; what Lean proves is the integer inequality `lemA`, with a weaker error term, directly). (ii) Everything in Section 8: the certificates at $L = 192$ and $L = 512$ are checked by Python only. (iii) The search for the certificates (floating point) is untrusted by design. (iv) The mathematics of the claim of Section 10 as a text: Theorem C is a Lean theorem whose statement is the upstream one, and the lanes' interface was audited against the claim's TeX source by an AI agent only (Sections 10.1 and 10.3); Theorems A and B do not depend on any of it. (v) That $\delta_3$ as defined is the constant the problem intends: this is the statement of the Formal Conjectures authors, and we only checked that it is the one we proved.

7.6. **Independent checks outside Lean.** Python scripts (`computation/theorem-b/`) rederive the kernel $K4$ from the closed form of Lemma 5.1 and compare it with brute-force cell counts of $R_n$ ($w = 2, 4, 6$); check each cut at all eight cube vertices; and prove positive definiteness of $N$ by Bareiss elimination ($L \leq 256$) or by exact diagonal dominance after an integer congruence ($L = 128, 256, 512$). The logs are in

`computation/theorem-b/logs/` and agree with the Lean statements for $L = 64, 128, 256$. `computation/small-cases/disc_cutcell_check.py` checks Lemma 5.4 exhaustively on sign vectors for $w \leq 10$ (largest excess 4; the lemma allows $4w$). The research log of the project reports further checks that are *not* shipped or re-run here: exhaustive enumeration of all colourings for tiny $(L, w)$, an exact worst-case check of the discrete cut-cell lemma for $w \leq 24$, and annealing of real colourings against the cover inequality. **Review.** No human has reviewed the prose of Sections 5–6; the AI reviews of it are listed in Appendix B.

7.7. **Standalone rebuild.** On 26 October 2026 the 189 modules of the package's Lean tree (by hash the same sources as `lean/` of the release repository) were compiled from the shipped sources, in dependency order, by plain `lean` 4.27.0 (not `lake`; `verification/build_closure.py`, which passes the options of `lakefile.toml`), two jobs at a time, against the compiled Mathlib of the research repository's checkout. Its `lake-manifest.json` lists the same package revisions as the shipped one (Mathlib `a3a10db0e9d6`). All 189 compilations returned 0; none was skipped as up to date. The run took 47 minutes of wall-clock time (08:42–09:29 PDT) and 92 minutes of summed compile time, on a machine shared with other jobs; the 76 modules of the $L = 256$ certificate took 51 of those 92 minutes. The axiom audit of Section 7 was then re-run with `LEAN_PATH` set to the rebuilt tree followed by the Mathlib packages only (no oleans of the research repository's own modules): exit code 0, 20 `#print axioms` lines, all `[propext, Classical.choice, Quot.sound]`. Logs: `earlier-runs/standalone_build_log.jsonl` (return code and seconds per module), `standalone_build_stdout.txt`, `lean_axiom_audit_shipped_tree.log`. The 153 further modules of Theorem C were rebuilt in the same way on the same day (Section 10.2). This run did not use `lake` and did not rebuild Mathlib. The run of record of the release repository (Section 7.8) adds a clean `lake` build of all 342 modules and a kernel replay, again against a compiled Mathlib that was copied, not rebuilt or downloaded. The package's compile options include `-Dwarn.sorry=false`, as in its project file, so its compile output is not a test for `sorry`; the axiom audit is. (The release repository does not set this option.)

7.8. **Run of record of the release repository.** The release repository [14] (Appendix A) holds the 342 modules of Theorems A–C, byte-identical to the research repository's at its merge commit 202f974, in a Lake project whose root module imports all of them, with a script, `verification/run_all.sh`, that repeats every formal and computational check from a clean build. On 26 October 2026 the script was run on a fresh clone of the repository's first commit (f68f061), with two jobs, on the same shared machine and through the same job gate, against a copy of the compiled Mathlib (a3a10db) of the research repository's checkout and without network access (15:59–22:05 PDT, 6 hours 6 minutes). It ended `RESULT: PASS`:

- **Build.** `lake build` of all 342 modules from scratch: `Build completed successfully (8228 jobs)`, in 147 minutes, with no `sorry` warning and 11 linter warnings (10 unused `simp` arguments, one deprecated lemma name).
- **Axioms.** `verification/Headline.lean`: 36 `#print axioms` lines, all `[propext, Classical.choice, Quot.sound]`. `verification/AxiomAudit.lean`: every one of the 4978 constants (2356 theorems) of the 342 modules depends only on these three axioms. The upstream pin, which copies the upstream statements of `prs_bounds`, `prs_conjecture`, `van_der_waerden` and `probabilistic` verbatim from [15] at c83a94d, proves them from ours and proves the definitions equal by `rfl`: 7 axiom lines, all standard.

- **Kernel replay.** `lean4checker` v4.27.0 [9] replayed each of the 342 modules in its own process, two at a time: all 342 exited with code 0 (3 hours 18 minutes; the longest module took 442 seconds).
- **Exact arithmetic and small cases.** All pass: the source scan, the statement-text check of Section 7, 14 checks of the numbers quoted in this paper, the regeneration of the 302 generated modules byte for byte from the JSON data, the certificates of Theorem B (18 runs) and of the five ranges of Theorem C (with Bareiss elimination), the value at the 12-block colouring, a comparison, at sample points, of the Lean definitions of the Theorem C interface with the Python model that made its certificates, the definition for $n \le 26$ against A121385, the cut-cell check and the symbolic lemma.

The largest resident memory of a single process was 8.7 GB for `lean` and 8.1 GB for `lean4checker`, close to the gate's 9 GB cap. Not re-run: the negative controls of Theorem B. One line of the logs, in which a script printed the local path of the `lean4checker` executable, was replaced after the run by what the corrected script prints; the logs (`verification/logs/`, summary in `SUMMARY.txt`) are otherwise as written. The repository's continuous-integration workflow, with one job and without the kernel replay, has never run.

## 8. Evidence beyond the formal results

All items in this section are computations *outside* Lean; none is used in Theorems A and B. The certified bounds are exact (rational certificates checked in exact integer arithmetic by Python) and are printed rounded *down*; everything that comes from a floating-point run is marked *(float)*.

| $L$ | cuts | certified bound | decimal (rounded down) | gap$\cdot L^2$ | in Lean |
|---|---|---|---|---|---|
| 32 | 127 | 109667/2097152 | 0.0522933 | 1.11 | no |
| 64 | 845 | 1782409/33554432 | 0.0531199 | 1.05 | yes |
| 128 | 3586 | 1788837/33554432 | 0.0533114 | 1.06 | yes |
| 192 | 5783 | 4027621/75497472 | 0.0533477 | 1.04 | no |
| 256 | 11931 | 14324135/268435456 | 0.0533615 | 0.94 | yes |
| 512 | 74264 | 28651667/536870912 | 0.0533678 | 2.10 (solver slack) | no |

Table 3. Shor + triangle-cut certificates for the cover of Section 5; gap = 117/2192 − bound. Positive definiteness: Bareiss elimination for $L \le 256$ and diagonal dominance for $L = 128, 256, 512$.

The gap law. For $L = 64, 128, 192, 256$ the gap to 117/2192 times $L^2$ is $1.05, 1.06, 1.04, 0.94$. At $L = 512$ the exact certificate has gap$\times L^2 = 2.10$, but there the augmented-Lagrangian dual had not converged (the floating-point primal value of the same relaxation is 0.0533720, the raw dual 0.0533151 (both float), the repaired and certified dual 0.0533678), so this is most likely solver slack and not a property of the model (we did not prove that). At $L = 1024$ (343,040 cuts) only a floating-point run exists: the primal value is 0.0533745 (float; a number, not a bound, since the cuts are not exactly satisfied) and the repaired dual is 0.0533491 (float), below the $L = 512$ certificate; no exact certificate was made. The smallest values of the cover's quadratic form over the cube found by floating-point multi-start local search are 0.0531202, 0.0533119, 0.0533617 (float; in the units of the bound $b(L, \sigma)$) at $L = 64, 128, 256$. Every cube point gives an upper bound for the best bound $b(L, \sigma)$ that this cover can yield, so these values are upper bounds for it (up to rounding), and the certified bounds lie below them by $2.7 \cdot 10^{-7}$, $4.0 \cdot 10^{-7}$, $1.4 \cdot 10^{-7}$. Hence the Shor-plus-triangle relaxation loses at most

about that much on this cover at the sizes tested (the local values are floating-point, so this statement is as reliable as they are, not exact). Together this is consistent with $\delta_3 = 117/2192$ and a cover whose loss is $\approx 1.0/L^2$. It does not prove it: a proof would need the minimum of the cover to converge to $117/2192$ *and* the relaxation to stay tight for all $L$, neither of which we know, and an equally good fit would hold if $\delta_3$ were slightly below $117/2192$ by less than the resolution of the data. (Section 10.1 gives a Lean proof of the equality, from a different argument, which, subject to the trust base of Section 7, settles the question; the evidence above was gathered independently of it and is not used in it.)

The strip cover with the same cuts. Keeping the strip cover of [11] but bounding the cube minimum by the same relaxation (with its own cut set) gives exactly certified bounds (Table 4); the sharper cover is higher at every size, by $1.7 \cdot 10^{-3}$ at $L = 64$ down to $2.4 \cdot 10^{-4}$ at $L = 512$.

| $L$ | cuts (strip) | strip + cuts, certified | strip cover, ceiling | sharper cover + cuts |
|---|---|---|---|---|
| 64 | 2167 | 0.0514278 | 0.0514282 | 0.0531199 |
| 128 | 4514 | 0.0523985 | 0.0523987 | 0.0533114 |
| 256 | 22066 | 0.0528846 | 0.0528851 | 0.0533615 |
| 512 | 101524 | 0.0531289 | 0.0531334 | 0.0533678 |
| 1024 | 363254 | 0.0532212 | 0.0532540 | (float only) |

TABLE 4. The strip cover of [11] with triangle cuts (certificates from the research repository, checked in Python only; exact positive definiteness by Bareiss for $L \le 256$ and by diagonal dominance; the $L = 1024$ certificate is not shipped). The ceiling is $\frac{1}{16} - \frac{1}{8L} - \frac{S^*}{8L^2}$ with $S^* = -x^{*\top} A x^*$ at an exact rational cube point $x^*$, as recorded in the research repository (not recomputed here): every bound of the form $\frac{1}{16} - \frac{1}{8L} - \frac{\sigma}{8L^2}$ with $\sigma$ a cube bound for $A$ (Proposition 3.2), whatever relaxation produced $\sigma$, is at most this. For comparison, diagonal-only exact strip certificates are 0.0511246, 0.0516249, 0.0518729 at $L = 128, 256, 512$.

Diagonal certificates on the sharper cover. 0.0522076, 0.0521973, 0.0521658, 0.0521505 at $L = 50, 64, 128, 256$ (exact certificates, rounded down; the $L = 64, 128, 256$ values are in `computation/theorem-b/logs/verify_model2_diag.log`, the $L = 50$ value is from the research log and not shipped). Floating-point optima of the same diagonal relaxation (float) rise to a flat maximum near $L = 50$ and then decrease slowly (0.052198, 0.052166, 0.052151, 0.052145 at $L = 64, 128, 256, 512$), so the diagonal relaxation of this cover does not approach $117/2192$ at any size we computed.

## 9. Prior figures and novelty

Table 5 lists the figures for $\delta_3$ that we know from other sources, and the main results of this paper, with their status (the evidence tables of Section 8 list the other certified values of this work). The searches behind it (web and arXiv searches during October 2026 for improvements of either bound, and the reading of the sources recorded in `references.bib`) found no improvement of (1) in print, no formal proof of the bounds in Formal Conjectures at c83a94d (where the statements are `sorry`), and, apart from the claim of Section 10, nothing newer on $\delta_3$ than [11], [8] and [13] (De Loera, Ventura, Wang and Wesley [5] compute optimal colourings of small intervals for $x + y = 2z$ and discuss the method of [11], without a new bound for $\delta_3$). (A lane of the research project searched the citing literature of [11] and [8] and extracted and searched the full texts of the papers it judged relevant, among them [8], [13], [18] and [3], for

the constant; the agent that wrote this manuscript read the sources named in Table 5 only to the extent stated in `references.bib`.) Other repositories of formal proofs were not searched systematically. The lane of the research project that proved Theorem C reports for 26 October 2026, on local snapshots, that none of the 660 open pull requests of Formal Conjectures mentions problem 1186, that the `lean-proofs` snapshot has no file for it, and that the database lists the problem as open with no formal solution; the refresh below repeated the first of these checks but not the second. A refresh on 26 October 2026 found no improvement of (1), no claim of $\delta_3 = 117/2192$ other than that of Section 10, and no aiXiv paper on the problem. The database's data file then listed problem 1186 as open, with the fields `formalized: yes` (the Formal Conjectures statement, since 2026-10-20) and `formal_status: unformalized` (no formal proof recorded). The upstream file had a single commit (4beab72, 2026-10-20), its statements were still `sorry` on the main branch, and no open pull request or issue there concerned the problem. None of this establishes that nothing exists. We therefore say only: *not found in the searches described*.

A 2026 arXiv preprint [18] concerns the Butler–Costello–Graham conjecture on whether every pattern admits a colouring beating the random one; the agent that wrote this manuscript read its abstract only (it states no value or bound for $\delta_3$), and a lane of the research project extracted its full text and searched it for the constant without finding an improvement of (1).

## 10. The claimed proof that $\delta_3 = 117/2192$, and Theorem C

**The first part of this section describes a claim by another author; its mathematics is not a result of this paper. Section 10.1 reports a Lean formalisation of it, Theorem C.** On 5 October 2026 a proof claim for Problem 1186 was submitted to the proof-claims thread of the problem's page [1] by Carlos Toledo (user carlos_toledo), who states that it was produced with substantial AI assistance from Claude and that they read and checked the mathematics line by line [17]. It asserts $\delta_3 = 117/2192$, i.e. that the 12-block colouring is optimal, so that the conjecture of [11] holds; it concerns $k = 3$ only. The claimant's paper (its Section 7) and report page also describe a model of $L$ equal cells in the variable $\varphi$, bounded by a Shor relaxation with triangle cuts, with $\delta_3 \geq 0.05324920$ at $L = 256$; that bound was accepted by the claimant's first verifier and, by the claim's own statement, not re-decided by its second. The claim uses the model to explain why discretising $\varphi$ cannot reach $117/2192$, and its proof does not use it. The claimant's report page carries a correction dated 20 October 2026, after an outside audit, withdrawing a remark that the bathtub bound is exact to first order; neither the claim's proof nor its formalisation uses that remark. The post on the problem's page states that the work is not peer reviewed and not formalised in Lean, that its certificates are checked by two independent programs in exact rational arithmetic (a statement of the page; we have not seen those programs run; the claim's report adds that the second program was written by an AI agent given only the mathematics), and that appearing on the page does not guarantee correctness.

As we understand the argument (from the page, from the claim's TeX source in the claimant's public repository as read by an AI agent of the project, and from the reports of three AI reviewers in the same research project; we did not read the PDF), it has four steps. (1) By the counting lemma of [11] (the formalisation uses instead the exact $\pm 1$ form of Lemma 2.2) the problem reduces to the infimum of a quadratic functional $Q$ over measurable $\varphi\colon [0,1] \to [-1,1]$, with $Q(\varphi^*) = -5/137$ for the 12-block profile $\varphi^*$, which gives $1/16 - 5/548 = 117/2192$. (2) Writing $\varphi = \varphi^*(1 - 2\sigma)$ expands the functional as a linear term $\int h\sigma$ with $h \geq 0$ vanishing exactly at the block edges, plus a quadratic term. (3) A bathtub (rearrangement) bound for the linear part (the claim's Lemma 2) and, in the proof of its Lemma 3, bounds for each pair

| Figure for $\delta_3$ | Source | Status |
|---|---|---|
| $\leq 1/16 = 0.0625$ | random colouring (probabilistic method, as stated in [1] and [15]) | classical; its Lean statement `probabilistic` is proved in the same development (not a claim of this paper) |
| $\geq 1675/32768 = 0.0511169\ldots$ | [11] | refereed; Lean: here |
| $\leq 117/2192 = 0.0533759\ldots$ | [11]; found independently in [3] | refereed; conjectured sharp; Lean: here |
| $\approx 0.052341$ (lower) | Sjöland, with O. Kivinen [13], Sec. 8 | numerical, degree-3 relaxation, no certificate ("work under progress") |
| 12-block colouring optimal among anti-symmetric colourings with $\leq 12$ blocks | [8] | refereed (Math. Comp. 2024); not a bound on $\delta_3$ |
| $\geq 0.0516249$, $0.0518729$ ($L = 256, 512$) | this work (PRS method) | exact; Lean for $L = 256$; not claimed as a record |
| $\geq 0.0532492$ ($L = 256$) | claim [17], its Section 7 (cell model in $\varphi$, Shor + triangle cuts; not used by its proof) | unrefereed; a reviewer reproduced the model's local-search minima to six digits, nobody here reproduced the certified bound; not Lean |
| $\geq 0.0533615$ | this work (Theorem B) | Lean; "not found in the searches described" |
| $= 117/2192$ | claim [17] | unrefereed; its author reports checking it line by line, and an outside audit led to a correction (2026-10-20); three AI reviewers (one: "undetermined, no gap found"; one rebuilt the finite problems from the claim's text and certified them with its own certificates, in its reading of the definitions, a third re-verified those; the proofs of the lemmas were not checked). Formalised in Lean as Theorem C of this package (Section 10.1); merged into the research repository's master branch and built there from source; no human referee |

TABLE 5. Figures for $\delta_3$. "Lean" means a proof against the Formal Conjectures definitions with standard axioms.

of cells for the quadratic part bound the functional below by a function of finitely many cell averages of $\sigma$, for *every* measurable $\sigma$ (no limiting argument). (4) Five finite exact certificates, on grids of 130 and 146 cells, cover the range of $m = \int \sigma \in [0, \frac{1}{2}]$ (the range $[0, \frac{1}{5}]$, where the claim proves $L \geq y^\top P y$ with $P$ positive definite, and four slabs with positive bounds; $m \leq \frac{1}{2}$ by the symmetry $\varphi \mapsto -\varphi$). Three read-only AI reviewers worked on it on 26 October 2026. The first re-derived steps (1)–(2) and tested the lemmas of step (3) by exact computations on random inputs, and concluded that the claim is undetermined end to end, with no gap found;

it did not reproduce the finite certificates. The second rebuilt the five finite problems of step (4) from the claim's text, produced exact certificates of its own whose bounds are at least the constants stated in the claim, and checked them in exact arithmetic. The third, working from the second one's definitions, re-verified those certificates with independent code. None of them ran or checked the claimant's own certificates. That is not a referee report: the proofs of the lemmas of step (3) were not checked as proofs, the finite inequalities were checked in the reviewers' reading of the definitions, and none of those checks was in Lean.

10.1. **Theorem C: a Lean proof of the claimed equality.** The research project formalised the claim in Lean, from its description and its own reading of the argument, in parts, against the same Lean definitions as in Section 7. The parts are stated as precise hypotheses in an interface file written by the project's coordinator, who also fixed the two cell grids, the constants and the function names; the AI agents that proved the parts used the interface as given. An audit of the interface against the claim's TeX source (by an AI agent of the project, 26 October 2026; the source of draft v0.2 in the claimant's public repository, read as text; the PDF was not read, and we could not confirm that this source is the version linked on 5 October) found that the Lean development follows the claim's argument step by step, in a discrete form at $n = 1096w$. The expansion is the discrete form of the claim's identity $E(\sigma) = \int h\sigma + Q(\varphi^*\sigma)$, with $H_x \approx 2nh(x/n)$. Part (b) below is the claim's Lemma 2 (the bathtub bound) with the claim's rule for the constants $a_i$, $b_i$. Part (c) is the bound for pairs of cells used in the proof of the claim's Lemma 3. The function $L$, the five ranges of $m$, the symmetry $\varphi \mapsto -\varphi$ and the two grids are the claim's; the grids agree point for point with those of the claimant's published certificates for the ranges $[\frac{1}{5}, \frac{3}{10}]$ and $[\frac{2}{5}, \frac{9}{20}]$ (the other three certificate files were not opened). The formalisation departs from the claim in five places: (i) the counting step uses the exact identity of Lemma 2.2 at $n = 1096w$ and Lemma 2.1, where the claim uses the counting lemma of [11] with step functions; (ii) the value $-\frac{5}{137}$ at the 12-block colouring comes from the cover inequality of Theorem 5.7, where the claim computes $Q(\varphi^*)$ exactly; (iii) only $\pm 1$ colourings are treated, where the claim's Theorem 2 covers every measurable $\varphi\colon [0,1] \to [-1,1]$; (iv) the five finite statements are $L \geq 0$, weaker than the claim's $L \geq B_j > 0$ on the four slabs and $L \geq y^\top P y$ with $P$ positive definite on $[0, \frac{1}{5}]$, so the claim's equality case and its corollary are not formalised; (v) the certificates are the project's own, not the claimant's (other multipliers and form sets, with 6817 and 1200 triangle forms where the claim has 2204 and 470, and exact positive semidefiniteness where the claim uses $S + \varepsilon I$). The claim's Lemma 1 is not formalised and is not needed. Theorem C is the Formal Conjectures statement about the $\liminf$; the claim's Theorem 1 also asserts that the limit exists, which is not formalised. The grids' constants and the integer certificates were generated by the project from a reviewer's reconstruction of the claim's finite problems (not from the claimant's own data, which we did not run), and the proofs of the parts, with their explicit error constants, are the lanes' own. The result is Theorem C.

**Theorem 10.1** (Theorem C)**.** $\delta_3 = 117/2192$*. In Lean this is the theorem* `Erdos1186Day.erdos_1186.variants.prs_conjecture`*, of type* $\delta\, 3 = 117/2192$*, for the definitions of Section 7, with no hypothesis;* `#print axioms` *gives* `[propext, Classical.choice, Quot.sound]`*.*

*Credit.* The argument behind Theorem C, in the form in which the project's reviewers and coordinator read it, is the claim's author's [17]; the Lean formalisation, with the proofs of its parts, is by AI agents of the research project; the claim is unrefereed. This paper contributes the checks described below, not the argument. The headers of the Lean files say: "Formalises the proof claimed on erdosproblems.com/1186 (Proof claims, 2026-10-05,

unrefereed); the mathematics is its author's." (This sentence is in 151 of the 153 files; the interface file `P1186Ex` has a variant of it, and the generated grid-data file `P1186ExData` mentions "the claimed proof" without the credit sentence; we did not edit the shipped sources.) The statement is the upstream one: Formal Conjectures lists it as `research open` and `sorry` at the pin [15]. If the Lean definitions express the constant of the problem (the Formal Conjectures authors' claim, Section 7), Theorem C answers the conjecture of [11]; we did not find another formal proof of it in the searches described in Section 9. Theorem B is a consequence of Theorem C.

*How the Lean proof is organised.* It works at the scale $n = 1096\,w$ for every $w \geq 1$ ($1096 = 2 \cdot 548$, so that the 12-block colouring $s^*$ of Section 4 is constant on each of 548 consecutive cells of $2w$ points). Let $R = R_n$ be the parallelogram of Lemma 2.2, $Q(u) = \sum_{(x,y)\in R} u_x u_y$ (so $W = Q(s)$), and write a colouring as $s = s^*(1 - 2\sigma)$ with $\sigma \in \{0,1\}$ the indicator of the points where it differs from $s^*$. Put $H_x = -s^*_x\big(\sum_{y:(x,y)\in R} s^*_y + \sum_{e:(e,x)\in R} s^*_e\big)$ (`H2` in Lean). Then, exactly in $\mathbb{Z}$ and for every $\sigma$ (steps (1) and (2) above),

$$W(s) = W(s^*) + 2\sum_x H_x\sigma_x + 4\,Q(s^*\sigma).$$

The parts are the following. Each has an error term at most linear in $w$, written $-Cw$ with an explicit constant, which does not matter in the limit.

(a) *Star.* $W(s^*) \geq -\frac{5}{137}n^2 - 3288\,w$. (The claim states $Q(\varphi^*) = -\frac{5}{137}$ as an exact computation; the route below is the present paper's.) It comes from the cover inequality of Theorem 5.7 (Lean `W_lower`, with 548 blocks of length $2w$): $s^*$ is constant on each block, so the cut-cell loss is 0, and what remains is a sum over pairs of blocks whose value, $-43840 = -\frac{5}{137}\cdot 1096^2$, the kernel evaluates after closed forms for the column sums of the cell kernel $K4$ (a sum of 274 terms). The constant $\frac{5}{137}$ gives $\frac{1}{16} - \frac{5}{548} = \frac{117}{2192}$.

(b) *Bathtub.* Two grids of cells are fixed, $G_{24}$ with 130 cells and $G_8$ with 146 cells, each cell $i$ with rational constants $a_i, b_i$ and a length $\omega_i$ (as a fraction of $[0,1]$). With $y_i$ the density of $\sigma$ on cell $i$, $\sum_{x\in\text{cell }i} H_x\sigma_x \;\geq\; 2n^2(a_i y_i + b_i y_i^2) - C_b w$ for every $\sigma$ and every cell. The proof uses that $H_x$ is within 2 of an explicit affine function of $x$ on each of the 1096 pieces of length $w$ (integer tables), a discrete rearrangement ("bathtub") lemma, and a Boolean check per cell by the kernel, in 5 chunks for $G_{24}$ and 4 for $G_8$.

(c) *Pairs.* For any well-formed grid, $Q(s^*\sigma) \geq n^2\sum_{i,j}\pi_{ij}(y_i, y_j) - C_p w$. With $P = \omega_i\omega_j y_i y_j$, $A^{\text{in}}_{ij}$ the area of the part of cell $i\times$ cell $j$ inside the counting region (computed exactly from the unit-cell kernel $K4$) and $A^{\text{out}}_{ij} = \omega_i\omega_j - A^{\text{in}}_{ij}$, the function $\pi_{ij}$ is $\max(0, P - A^{\text{out}}_{ij})$ if $s^*$ has the same sign on the two cells and $\max(-A^{\text{in}}_{ij}, -P)$ otherwise.

(d) *Finite certificates.* Put $L(y) = \sum_i(a_i y_i + b_i y_i^2) + \sum_{i,j}\pi_{ij}(y_i, y_j)$ for $y \in [0,1]^N$, and $m = \sum_i \omega_i y_i$. There are five statements, $L(y) \geq 0$ for $m$ in $[0, \frac{1}{5}]$, $[\frac{1}{5}, \frac{3}{10}]$, $[\frac{3}{10}, \frac{2}{5}]$ (on $G_{24}$) and $[\frac{2}{5}, \frac{9}{20}]$, $[\frac{9}{20}, \frac{1}{2}]$ (on $G_8$); the claim proves the stronger bounds $L \geq B_j > 0$ on the four slabs and $L \geq y^\top P y$ on $[0, \frac{1}{5}]$, but only $L \geq 0$ is needed, and only that is formalised. Each is proved by a certificate of the type of Section 6: $L(y)$ is bounded below by a quadratic form with an exact integer matrix plus nonnegative multiples of valid inequalities (box constraints and, for the ranges that need them, triangle forms: 6817 in the third range, 1200 in the fifth, none in the others), and positive semidefiniteness of the matrix is shown by an approximate Cholesky factor with diagonally dominant remainder. The integer data were produced by another agent in Python, from a reviewer's reconstruction of the finite problems, and are checked in

Lean by `decide +kernel` on packed integer tables, the triangle sums as exact integer sums in chunks. The soundness library of this format contains no numbers.

(e) *Assembly.* Adding (a), twice (b) summed over the cells and four times (c) gives $W(s) \geq -\frac{5}{137}n^2 + 4n^2L(y) - Cw$. A density $m \in [0, \frac{1}{2}]$ falls in one of the five ranges; the symmetry $\sigma \mapsto 1-\sigma$ (that is, $s \mapsto -s$, which leaves $W$ unchanged) covers $m > \frac{1}{2}$. Then $32V \geq 2n^2 - 16n + 8W$ (Lemma 2.2) gives $V(n,c) \geq \frac{117}{2192}n^2 - O(n)$ at $n = 1096w$, the passage to all $n$ is as in Lemma 2.1, and the upper bound of Theorem A closes the equality.

| Module(s) | Content |
|---|---|
| `P1186Ex`, `P1186ExData` | the interface: scale $n = 1096w$, the colouring $s^*$, the two grids and their constants, the statements of (a)–(d) |
| `P1186ExStar` | (a) |
| `P1186ExBath*` (16 modules) | (b) on $G_{24}$ and $G_8$: the affine approximation of $H_x$, the discrete bathtub lemma, per-cell checks in chunks |
| `P1186ExPair` | (c) |
| `P1186ExGlue`, `P1186ExMain` | (e), and the theorem `delta3_eq_of_regions`: $\delta_3 = \frac{117}{2192}$ from the five statements of (d) as hypotheses |
| `P1186ExCertLib`, `-Alg`, `-Chk`, `-Tri`, `-Pk` | soundness of the certificate format (hand-written, no numbers) |
| `P1186ExCert1`–`5` (19, 19, 33, 23, 30 modules) | (d): generated tables and chunked kernel checks, one family per range |
| `P1186ExFinal` | Theorem C |
| `P1186ExCertCtl` | negative controls (Section 10.2) |

TABLE 6. The 153 Lean modules behind Theorem C, beyond the 189 of Theorems A and B (342 in all). The closure of `P1186ExFinal` has 163 modules: these 152 (all but `P1186ExCertCtl`) and 11 shared with the closure of Theorem A or B (`P1186`, `P1186Cert128Lib`, `P1186Low`, `P1186LowA`, `P1186M2`, `P1186M2CertLib`, `P1186M2Low`, `P1186M2LowA`, `P1186M2LowB`, `P1186Prs`, `P1186Tab`). All 342 are in `lean/Erdos/Proofs/` of the release repository.

10.2. **Checks of Theorem C.**

- **Which sources.** The 153 modules were copied, as LF text, from the git blobs of commit f726409 of the branch `agent/n3c` of the research repository, the tip of the formalising lane at the time (the modules `P1186ExFinal` and `P1186ExCertCtl` were added by the commit just before, 97b5bfa; f726409 changed the import list and notes; the two later commits, b451885 and a9b4235, differ from it in two documentation files and in no Lean source, as we checked by a read-only `git diff` at 14:18 PDT). The 11 modules shared with Theorems A and B are byte-identical in the two commits that supply the package.
- **Statement.** `verification/fidelity_check.py`, given `P1186ExFinal.lean` as a fifth argument, checks that Theorem C has the name, the type and the absence of binders of the upstream statement, and the namespace of the development (`Erdos1186Day`, as for the definitions); the log ends `RESULT: ALL OK`. The definitions are those of Section 7.
- **Source scan.** The closure of `P1186ExFinal` (163 modules, 8,943,323 bytes) has 11 source lines containing one of the forbidden words (log

earlier-runs/closure_scan_claim_tree.log); all are comments saying "no native_decide". P1186ExCertCtl has none.

- **Standalone rebuild.** On 26 October 2026 the same script as in Section 7.7 (plain lean 4.27.0, not lake; two jobs; the same compiled Mathlib) was run on the roots of Theorems A, B and C: 341 modules, of which the 189 already compiled in the earlier run (Section 7.7, same sources by hash) were skipped as up to date and 152 were compiled from the shipped sources. All 152 compilations returned 0. The run took 87 minutes of wall-clock time (12:36–14:04 PDT) and 174 minutes of summed compile time, on a machine shared with other jobs, including the coordinator's own gated build in the research repository. The last module built was P1186ExFinal. The 153rd module, P1186ExCertCtl, was compiled separately (below). Logs: standalone_build_claim_log.jsonl, standalone_build_claim_stdout.txt.
- **Axioms.** earlier-runs/AuditClaim1186.lean was run with LEAN_PATH set to the rebuilt tree followed by the Mathlib packages only: exit code 0; it prints the definitions of monoCount, minMonoCount and $\delta$, the statement of Theorem C and of delta3_eq_of_regions, and 15 #print axioms lines, for Theorem C, delta3_eq_of_regions, delta3_eq_of_parts, lower_of_parts, starOK, bathOK24, bathOK8, pairOK_of_WF, pairOK24, pairOK8 and the five region theorems; all give [propext, Classical.choice, Quot.sound] (log lean_axiom_audit_claim_tree.log). The compile options include -Dwarn.sorry=false; this audit, not the compile, is the test for sorry.
- **Negative controls.** P1186ExCertCtl (region 5, one of the two ranges with triangle cuts, grid $G_8$) states, as kernel-checked examples, that the checkers accept the true packed constant, a linear entry and a pair entry, and reject the constant plus one, a linear entry plus $2^{7W}$ ($W$ the digit width of the packing) and the quadratic table with 1 added to each entry; that a dominance row is accepted for the true Cholesky data and rejected when one Cholesky entry is changed by $10^{12}$; and that the exact group sums of the constant, linear and quadratic parts differ from the claimed values plus one. (Changing that Cholesky entry by 1 is *not* rejected: the row has margin, and a certificate needs only some valid factor.) These controls cover region 5 only; P1186ExBath.lean contains nine further corruption examples for the bathtub check (report of that lane, not re-examined), and none was shipped for ranges 1–4. We compiled P1186ExCertCtl from the shipped source (one job, 74 seconds, return code 0; lean_build_controls.log), so the kernel accepted all 11 examples.
- **Kernel replay.** Two first attempts, on the package, to replay the closure of Theorem C (163 modules; the replay script listed 164) with lean4checker (built from tag v4.27.0 in the tools of another package of this project; chunks of eight modules, through the machine's job gate) did not get past the first chunk: the first stopped after 415 seconds with "failed to read file" on a Mathlib .olean that exists on disk (not diagnosed), and the second was killed after 64 seconds (exit code 137; most likely the gate's 9 GB memory cap). Logs: lean4checker_claim_attempt1_failed.log, lean4checker_claim_attempt2_killed.log. The run of record of the release repository (Section 7.8) then replayed all 342 modules, those of Theorems A and B included, one process per module and two at a time: all 342 exited with code 0.
- **Build in the research repository.** The branch was merged into the research repository's master branch on 26 October 2026 (merge commit 202f974; its Lean sources for Theorems A–C are byte-identical to the shipped ones). Before the build, the 124 generated certificate modules were regenerated from the integer data with the generator

and compared with the merged files: all identical. Then all 131 certificate-side modules were built from source with `lake`, one gated `lake build` per module in dependency order (13:24–15:13 PDT, 111 minutes of summed build time; every build successful); the 22 other modules of the closure (interface, star, bathtub, pair, assembly) had been built from source with `lake` in the master branch as their parts were merged. There `#print axioms` gives `[propext, Classical.choice, Quot.sound]` for Theorem C, the five region theorems and `delta3_eq_of_regions`, and the repository's verification tool reports 861 proved declarations for the problem and none with `sorry`.

The formalising lane's own report (not re-run here) adds: gated builds of every module; a Python re-derivation of every identity of the generated tables for all five ranges; and its own listing of the `#print axioms` of the five region theorems and of Theorem C, all standard.

10.3. **Limits of Theorem C.**

- It is an unrefereed claim's argument, formalised by AI agents in one day. So far as we know, apart from the claimant, who states that they read and checked the mathematics line by line, and the outside audit recorded in the claim's report (auditor not named), no human has read the claim's argument, and no human has read the Lean sources beyond the statement of Theorem C.
- What makes Theorem C true is that the kernel accepts the upstream statement with standard axioms. The trust base is that of Section 7: Lean's kernel, Mathlib at the pinned version, and the faithfulness of the upstream definitions to the problem. Whether $\delta_3$ so defined is the constant the problem intends is the Formal Conjectures authors' claim.
- The interface of the lanes (the grids $G_{24}$, $G_8$, the constants $a_i, b_i$, $H_x$, $\pi_{ij}$) was audited against the claim's TeX source by an AI agent of the project, not by a human and not against the PDF (Section 10.1); the claimant's values of $a_i, b_i$ (which their certificates do not store) were not compared with ours, only the rule that produces them. The claim, the reviewers, the formalising agents and the auditor all used Claude models, so these checks cannot exclude a shared misreading of the claim. This does not affect the truth of Theorem C.
- All the machine checks of Theorem C, in the research repository and here, were run by AI agents of the project on one machine; none was repeated by a person or on another machine.
- The Python programs that produced the integer tables are not part of the proof; the tables are re-checked by the kernel.

10.4. **History of the formalisation.** The formalisation was done on 26 October 2026, in parts. By about 09:30 PDT (commit b23e2f8 of the research repository) the assembly and the star, bathtub and pair parts were proved and merged, and `delta3_eq_of_regions` gave $\delta_3 = 117/2192$ with the five finite statements as hypotheses; at that time those five statements were not in Lean (their integer certificates existed and were checked by Python only). The lane writing the Lean checker committed the proofs of the five statements and `P1186ExFinal` on its branch (commit 97b5bfa, 12:06 PDT; the tip f726409 followed a minute later). The lane reported itself done at about 13:20 PDT. The project's coordinator merged the branch into master, built it there from source and recorded the result between 13:24 and 15:25 PDT (merge commit 202f974). An earlier state of this manuscript reported only the conditional theorem and did not claim Theorem C.

## 11. Discussion and limitations

- **What Theorem B is now.** Theorem B narrows the interval for $\delta_3$ to relative width $2.7 \cdot 10^{-4}$. Theorem C, subject to the trust base of Section 7, gives the equality and makes Theorem B a consequence of it; Theorem B keeps its value as a different route: a cell cover of the whole counting region with a single relaxation, whose proof does not use the claim's argument (the expansion around the optimal profile), although both end in semidefinite certificates with triangle inequalities.
- **What Theorem C is.** A Lean proof, by AI agents, of the equality claimed in another author's unrefereed argument. Its truth rests on the kernel and the upstream definitions; the mathematics is not ours, so far as we know no human has read the Lean text, and the lanes' interface was audited against the claim's TeX source by an AI agent only (Section 10.3).
- **Trust base.** Lean's kernel, Mathlib at the pinned version, and the faithfulness of the upstream definitions. The Python scripts are cross-checks, not part of the proof.
- **Scope.** A proof of the displayed Lean statements is a proof of bounds for the lim inf as formalised, for $k = 3$. It says nothing about Problem 1186 as posed for general $k$.
- **Scaling.** The kernel scheme costs about $L^3/3$ multiply-adds, so each doubling of $L$ multiplies the kernel time by about 8; $L = 256$ already needs 76 modules and, in summed compile time, 51 minutes (the standalone rebuild, Section 7.7) or 1.5 hours (the formalising agents' gated run); the two figures were measured under different conditions and are not comparable. With a gap of about $1/L^2$, this method alone reaches 117/2192 only in the limit, and every finite $L$ stays below it.
- **Provenance.** The cover, the lemmas and the certificates were generated by AI agents. The Lean statements and certificates were checked by machine; the prose proofs of the lemmas of Section 5 were not formalised and were not checked by a human mathematician (Appendix B). The mathematics behind Theorem C is another author's.

## Appendix A. Reproducibility

**Release repository.** The Lean sources, the certificates, the scripts and the logs form the release repository [14], version 1.0.0. `lean/` is a standalone Lake project (Lean v4.27.0, Mathlib a3a10db, tag v4.27.0) with the 342 modules under `Erdos/Proofs/`, byte-identical to the research repository's at 202f974 (the paper's package took the 189 modules of Theorems A and B from its commit 47bee1f and the 153 of Theorem C from commit f726409 of the branch `agent/n3c`, Section 10.2), and a root module `Erdos1186.lean` whose import closure is all 342. `verification/` holds `run_all.sh`, `Headline.lean`, `AxiomAudit.lean`, the upstream pin (`make_upstream_pin.py` and the pinned upstream file), `source_scan.py`, `fidelity_check.py`, `regen_check.py`, `number_checks.py`, `build_closure.py` (the module-by-module build with plain `lean` of Section 7.7), the logs of the run of record (`logs/`) and the files and logs of the earlier runs on the paper's package (`earlier-runs/2026-10-26-package/`, written `earlier-runs/` in the text; the logs named without a directory in Sections 7.7 and 10.2 are there). `computation/` holds the certificates (JSON), their generators and exact verifiers for Theorem B (`theorem-b/`) and Theorem C (`theorem-c/`), and the small-case checks (`small-cases/`); `docs/` has the provenance, a guide for a human checker, the literature search and the internal reviews; `MANIFEST.sha256` lists the SHA-256 of every file.

**Commands.** From the repository root, `FETCH_CACHE=1 JOBS=2 bash verification/run_all.sh` downloads Mathlib and its build cache, builds the 342 modules from scratch, runs every check of Section 7.8 and writes `verification/logs/SUMMARY.txt`;

with `LEAN4CHECKER` set to a `lean4checker` v4.27.0 executable it also replays every module. It needs about 9 GB of memory per job and several hours. `bash computation/theorem-b/run_checks.sh quick` (about 3 minutes) or `full` (adds $L = 512$; about 13 minutes in all) re-runs the Python verifications of the certificates of Theorem B and of Section 8 alone; the repository's README describes the other scripts.

**Code and data availability.** The release repository (code under the Apache License 2.0, text and data under CC BY 4.0) is not yet public; it is available from the author on request.

## Appendix B. Statement on the use of AI

The mathematical content of Sections 5–8 (the sharper cover, the cut-cell lemma, the choice and certification of the relaxation, the certificates), the Lean formalisation, the scripts and the text of this manuscript were produced by AI systems (Claude models from Anthropic, running as autonomous agents in a research workspace, on 23–26 October 2026), each task run by a separate agent with its own written report (a *lane* in the project's records and in this paper; builds ran under a job gate that limits each process tree to 9 GB of memory). The research workspace was set up and directed by the author, who chose the problem, commissioned the runs and decided what to publish. A coordinating agent (Claude Opus 5.5) wrote the task briefs, including the hypothesis that led to the cover of Section 5, wrote the interface of Theorem C (`P1186Ex`, `P1186ExData`) and its assembly `P1186ExMain`, and merged and built the agents' branches. Most task agents, including the one that drafted this manuscript, ran Claude Sonnet 5.5; a few ran Claude Opus 5.5. The AI referees of this manuscript, the three reviewers of the claim and the auditor of the formalisation were also Claude agents. The formal proofs use Lean 4.27.0 and Mathlib. By its author's statement, the claim formalised in Theorem C was produced with substantial AI assistance (Claude). The Lean proofs were checked by the Lean kernel and the scripts' results compared across independent implementations; further AI agents reviewed the claim discussed in Section 10, without a human referee, and other AI agents formalised it (Theorem C, Section 10.1). The mathematics of that claim, and of [11], is its authors'; the Lean proofs and the checks of Section 10.2 were produced by AI agents. The project's records attribute the mathematics of Theorem B, all the formal proofs and the text to these AI systems and record no human mathematical contribution. The results have not been checked by an independent human mathematician or refereed; they are supported by the formal verification of Sections 7 and 10.2. *Reviews.* One fresh AI referee (a Claude Opus agent, 26 October 2026) tried to break the reduction, the cell geometry, the cut-cell lemma and the cover inequality of the research note on which Sections 5–6 are based and found no error; it did not check any certificate or table. Four fresh Claude agents refereed this manuscript on 26 October 2026: three read successive drafts (the third the draft that contained Theorem C) and a fourth the near-final text; two further agents audited the description of the claim against its source and the references. Their findings are incorporated. No human has read the manuscript. The author takes full responsibility for the content of this paper, including the accuracy of this statement, and for correcting or withdrawing any result that is found to be wrong.

## References

1. Thomas F. Bloom, *Erdős problem 1186*, `https://www.erdosproblems.com/1186`, last edited 8 April 2026, accessed 26 October 2026.
2. Steve Butler, *A121385: Minimal number of monochromatic three-term arithmetic progressions that a two-coloring of* $\{1, \dots, n\}$ *can contain*, The On-Line Encyclopedia of Integer Sequences, `https://oeis.org/A121385`, 2006.
3. Steve Butler, Kevin P. Costello, and Ron Graham, *Finding patterns avoiding many monochromatic constellations*, Experiment. Math. **19** (2010), no. 4, 399–411.

4. Steve Butler, Ron Graham, and Linyuan Lu, *Unrolling residues to avoid progressions*, Math. Mag. **87** (2014), no. 2, 83–94.
5. Jesús A. De Loera, Denae Ventura, Liuyue Wang, and William J. Wesley, *Optimization tools for computing colorings of* $[1, \dots, n]$ *with few monochromatic solutions on 3-variable linear equations*, Discrete Appl. Math. **373** (2025), 159–178, arXiv:2410.21651.
6. Leonardo de Moura and Sebastian Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction – CADE 28, Lecture Notes in Comput. Sci., vol. 12699, 2021, pp. 625–635.
7. Paul Erdős, *A survey of problems in combinatorial number theory*, Ann. Discrete Math. **6** (1980), 89–115.
8. Torin Greenwood, Jonathan Kariv, and Noah Williams, *From discrete to continuous: Monochromatic 3-term arithmetic progressions*, Math. Comp. **93** (2024), no. 350, 2959–2983, arXiv:2301.00336.
9. Lean FRO, *lean4checker*, `https://github.com/leanprover/lean4checker`, Tag `v4.27.0`.
10. Manfred Padberg, *The Boolean quadric polytope: some characteristics, facets and relatives*, Math. Programming **45** (1989), 139–172.
11. Pablo A. Parrilo, Aaron Robertson, and Dan Saracino, *On the asymptotic minimum number of monochromatic 3-term arithmetic progressions*, J. Combin. Theory Ser. A **115** (2008), no. 1, 185–192, arXiv:math/0609532.
12. N. Z. Shor, *Quadratic optimization problems*, Izv. Akad. Nauk SSSR Tekhn. Kibernet. (1987), no. 1, 128–139, English translation: Soviet J. Comput. Systems Sci. 25 (1987), 1–11.
13. Erik Sjöland, *Enumeration of monochromatic three term arithmetic progressions in two-colorings of any finite group*, arXiv:1408.1088v3, 2014.
14. Chandler Snively, *Erdős Problem 1186,* $k = 3$*: Lean formalisation, verification scripts and computations*, 2026, Release v1.0.0, prepared 26 October 2026; not yet public, available from the author on request.
15. The Formal Conjectures Authors, *Formal conjectures*, `https://github.com/google-deepmind/formal-conjectures`, file `FormalConjectures/ErdosProblems/1186.lean`, commit c83a94d785d0fb3b9b9a2a94e8ffff8cce76a346 (2026-10-22).
16. The mathlib Community, *The Lean mathematical library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), 2020, pp. 367–381.
17. Carlos Toledo, *The twelve-block colouring is optimal:* $\delta_3 = 117/2192$, Submitted to `https://www.erdosproblems.com/forum/thread/1186/proof-claims` on 2026-10-05 (user carlos_toledo); PDF at `https://carlostoledo.co/paper/delta3.pdf`; certificates and records at `https://carlostoledo.co/reports/delta3.html`; source at `https://github.com/carlostoledo1891/cert-machine`; archive doi:10.5281/zenodo.23171167 (the software record cert-machine v2026.10, as given in the claim; later archived versions doi:10.5281/zenodo.23446020 and doi:10.5281/zenodo.23551746); accessed 2026-10-26, 2026, Unrefereed proof claim, draft v0.2, October 2026.
18. Gang Yang and Yaping Mao, *Complete resolution of the Butler–Costello–Graham conjecture on monochromatic constellations*, arXiv:2604.02115, 2026.

Quana Labs
*Email address*: `lab@quanalabs.com`